

CMP020L006: Cyber Security

In this lab, you are going to build a sandboxed network using Virtual Box and 4 virtual machines. A network diagram of what you will deploy is shown in Figure 1.

How will the sandboxed network be used?

You will use this sandboxed network throughout the module for practical labs, some of which count as contributions towards your coursework(worth 60% of the total module marks).

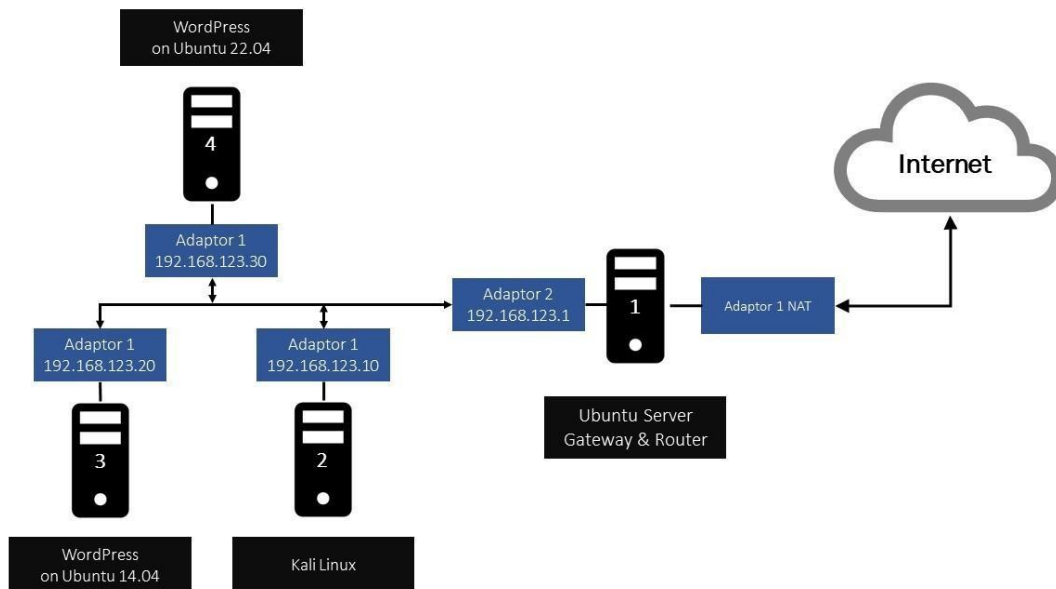


Figure 1: Sandboxed Network Diagram

Resources for this Lab

This lab requires 4 virtual machines. Details for each machine are summarised in Table 1, Table 2, Table 3 and Table 4 respectively.

Virtual Machine: 1	Ubuntu Gateway & Router
Use case(s)	- As an internet access gateway for WordPress and Kali virtual machines. - As a network Firewall
RAM	2 GB
CPUs	1
Virtual Machine: 1	Ubuntu Gateway & Router
Operating System	Ubuntu 22.04
Network Adaptor 1	NAT
Network Adaptor 2	192.168.123.1
Source	Cyber Lab VM Repository
Username	student
Password	Student1

Table 1: Ubuntu Gateway & Router

Virtual Machine: 2	Kali Linux
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Use case(s)	Workstation
RAM	4 GB
CPUs	2
Operating System	Debian
Network Adaptor 1	192.168.123.10
Source	Cyber Lab VM Repository
Username	kali
Password	kali

Table 2: Kali Linux

Virtual Machine: 3	WordPress
Use case(s)	Web Server (Content Management System)
RAM	2 GB
Virtual Machine: 3	WordPress (Legacy)
CPUs	1
Operating System	Ubuntu 14.04
Network Adaptor 1	192.168.123.20
Source	Cyber Lab VM Repository
Username	student
Password	Student1

Table 3: WordPress on Ubuntu 14.04

Virtual Machine: 4	WordPress (New)
Use case(s)	Web Server (Content Management System)
RAM	2 GB
CPUs	1
Operating System	Ubuntu 22.04
Network Adaptor 1	192.168.123.30
Source	Cyber Lab VM Repository
Username	student
Password	Student1

Table 4: WordPress on Ubuntu 14.04

Using your own Laptop or PC

You will need to download:

- Windows: [VirtualBox for Windows](#)
- Mac (Intel): [VirtualBox for Mac \(Intel\)](#)
- Mac (Silicon): [VirtualBox for Mac \(Intel\)](#)
- Linux: [VirtualBox for Linux](#)

If you plan to run this lab on your own laptop, you will need to have a **minimum of 16 GB of RAM**.

Lab Requirements

1. Copy the virtual machine .OVA files to your lab workstation or laptop

The virtual machines for this are available from the cyber lab VM repository.

2. Configure the network adaptors to use the IP addresses specified in Table 1, Table 2, Table 3 and Table 4 respectively.

If you are not familiar with configuring IP address for Linux virtual machines, research the process. However, we will cover this in the lab session.

3. Test that machines can communicate with each other. For example:

- Use the ping command and test for appropriate responses
- Use a browser to connect to WordPress
- Find the WordPress Admin URL and sign on

4. Test to see if all virtual machines have internet access.

- You can test communications using the ping command.
- Use a browser from your Kali Linux virtual machine.
- Use the ping command from servers (e.g., try pinging 8.8.8.8).

End of Lab :-)